

- 1b,3a,5b,11b,8b DHT 11 Sensor - Temperature and Humidity Readings

```
#include <Adafruit_Sensor.h> // include adafruit library
#include <DHT.h>
#include <DHT_U.h>
#define DHTPIN 32 // digital pin 32
#define DHTTYPE DHT11
DHT_Unified dht(DHTPIN, DHTTYPE);
void setup() {
  Serial.begin(115200);
  dht.begin();
}
void loop() {
  sensors_event_t event; // It's a class under that give name event
  dht.temperature().getEvent(&event); // dht is object,& address of object
  Serial.print("Temperature: "); // print temp
  Serial.print(event.temperature); // under event it displays temp
  Serial.println("°C");
  dht.humidity().getEvent(&event);
  Serial.print("Relative Humidity: ");
  Serial.print(event.relative_humidity); // serial is object
  Serial.println("%");
  delay(2000);
}
```

- 2b,4a,6b,10a OLED - Temperature and Humidity Readings

```
#include <Arduino.h>
#include <U8x8lib.h>
#ifdef U8X8_HAVE_HW_SPI
#include <SPI.h>
#else
#include <SoftwareSerial.h>
#endif
U8X8_SH1106_128X64_NONAME_HW_12C u8x8(/* reset=*/
U8X8_PIN_NONE);
#include <Adafruit_Sensor.h>
#include <DHT.h>
#include <DHT_U.h>
#define DHTPIN 32
#define DHTTYPE DHT11
DHT_Unified dht(DHTPIN, DHTTYPE);
void setup(void)
{
  u8x8.begin();
  u8x8.setPowerSave(0);
```

```

u8x8.setFont(u8x8_font_chroma48medium8_r);
dht.begin();
}
void loop(void)
{
u8x8.setCursor(0,1);
sensors_event_t event;
// get temperature and print as float
dht.temperature().getEvent(&event);
u8x8.print("Temp:");
u8x8.print(event.temperature);
u8x8.print(" Cels");
// get humidity and print as integer
u8x8.setCursor(0,3);
dht.humidity().getEvent(&event);
u8x8.print("Hum:");
u8x8.print(event.relative_humidity);
u8x8.print("%");
delay(2000);
}

```

- 1a,7b,8a,11a LED/Buzzer - Turn On LED for 1 second after every 2 seconds

```

void setup() {
pinMode(2, OUTPUT); // Set GPIO2 as output
}
void loop() {
digitalWrite(2, HIGH); // Turn LED ON
delay(1000);
// Wait 1 second
digitalWrite(2, LOW); // Turn LED OFF
delay(2000);
// Wait 2 seconds
}

```

- 2a,9b Push Button/Digital Sensor (IR/LDR)

```

void setup() {
pinMode(18, OUTPUT); // Set GPIO18 as output for LED
pinMode(35, INPUT); // Set GPIO35 as input for Push button or Sensor
}
void loop() {
int val;
val = digitalRead(35); // Read push button or sensor state

```

```

if (val = HIGH) {
digitalWrite(18, HIGH); // Turn LED ON if button is pressed or object
} else {
digitalWrite(18, LOW); // Turn LED OFF otherwise
}
}

```

- 5a,10b Motor - Turn On when push button is pressed

```

// To interface motor using relay with Arduino/Raspberry
// Pi and write a program to 'turn ON' motor when
// push button is pressed.
void setup() {
// put your setup code here, to run once:
pinMode(13,OUTPUT);
pinMode(35,INPUT);
}
void loop() {
// put your main code here, to run repeatedly:
if(digitalRead(35))
{
digitalWrite(13,HIGH);
}
else
{
digitalWrite(13, LOW);
}
}

```

- 4b,9a Thingspeak Cloud (Retrieve)

```

#include <Adafruit_Sensor.h>
#include <DHT.h>
#include <DHT_U.h>
#define DHTPIN 32
#define DHTTYPE DHT11
DHT_Unified dht(DHTPIN, DHTTYPE);
#include <WiFi.h>
#include "ThingSpeak.h"
const char* ssid = "Namratha";// your network SSID (name)
const char* password = "kruthika";// your network password
WiFiClient client;
unsigned long myChannelNumber = 3;
const char myWriteAPIKey = "M62TBVDF7LA9F203";

```

```

const char* readAPIKey = "P261LAD39SDRKP2X";
const long channelId=2456613;
const unsigned int firstReadFieldNumber = 1;
const unsigned int secondReadFieldNumber = 2;
const unsigned int switchField = 3; // Field number (1-8) to use to change status of device.
Determines if data is read from ThingSpeak.
// Timer variables
unsigned long lastTime = 0;
unsigned long timerDelay = 30000;
float temperatureC;
float humidityP;
float temperatureR;
float humidityR;
void setup() {
  Serial.begin(115200); //Initialize serial
  dht.begin();
  WiFi.mode(WIFI_STA);
  ThingSpeak.begin(client); // Initialize ThingSpeak
}
void loop() {
  if ((millis() - lastTime) > timerDelay) {
    // Connect or reconnect to WiFi
    if(WiFi.status() != WL_CONNECTED){
      Serial.print("Attempting to connect");
      while(WiFi.status() != WL_CONNECTED) {
        WiFi.begin(ssid, password);
        delay(5000);
      }
      Serial.println("\nConnected.");
    }
    sensors_event_t event;
    dht.temperature().getEvent(&event);
    temperatureC=event.temperature;
    dht.humidity().getEvent(&event);
    humidityP=event.relative_humidity;
    Serial.print("Temperature: ");
    Serial.print(temperatureC);
    Serial.println("°C");
    Serial.print("Relative Humidity: ");
    Serial.print(humidityP);
    Serial.println("%");
    // set the fields with the values
    ThingSpeak.setField(1,temperatureC);
    ThingSpeak.setField(2,humidityP);
  }
}

```

```

// Write to ThingSpeak. There are up to 8 fields in a channel, allowing you to store up to 8
different
// pieces of information in a channel. Here, we write to field 1.
int x = ThingSpeak.writeFields(myChannel Number, my WriteAPIKey);
if(x = 200) {
-Serial.println("Channel update successful.");
}
else{
Serial.println("Problem updating channel. HTTP error code " + String(x));
}
temperatureR=
ThingSpeak.readFloatField(channelID, firstReadFieldNumber, readAPIKey);
Serial.println(" Temperature read from ThingSpeak "+String(temperatureR));
humidityR=
ThingSpeak.readFloatField(channelID,secondReadFieldNumber,readAPIKey);
Serial.println(" Humidity read from ThingSpeak "+String(humidityR));
last Time = millis();
}
}

```

- 3b ThingSpeak Cloud (Upload)

```

#include <Adafruit_Sensor.h>
#include <DHT.h>
#include <DHT_U.h>
#define DHTPIN 32
#define DHTTYPE DHT11
DHT_Unified dht(DHTPIN, DHTTYPE);
#include <WiFi.h>
#include "ThingSpeak.h"
const char* ssid = "Namratha";// your network SSID (name)
const char* password = "kruthika";// your network password
WiFiClient client;
unsigned long myChannelNumber = 2;
const char *my WriteAPIKey = "KM0DAH7DAEWSL9P8";
// Timer variables
unsigned long lastTime = 0;
unsigned long timerDelay = 30000;
float temperatureC;
float humidityP;
void setup() {
Serial.begin(115200); //Initialize serial
dht.begin();
WiFi.mode(WIFI_STA);

```

```

ThingSpeak.begin(client); // Initialize ThingSpeak
}
void loop(){
  if ((millis()-lastTime) > timerDelay) (
    //Connect or reconnect to WiFi
    if(WiFi.status() != WL_CONNECTED){
      Serial.print("Attempting to connect");
      while(WiFi.status() != WL_CONNECTED) {
        WiFi.begin(ssid, password);
        delay(5000);
        Serial.println("\nConnected.");
        sensors_event_t event;
        dht.temperature().getEvent(&event);
        temperatureC=event.temperature;
        dht.humidity().getEvent(&event);
        humidityP=event.relative_humidity;
        Serial.print("Temperature: ");
        Serial.print(temperatureC);
        Serial.println("°C");
        Serial.print("Relative Humidity: ");
        Serial.print(humidityP);
        Serial.println("%");
        // set the fields with the values
        ThingSpeak.setField(1,temperatureC);
        ThingSpeak.setField(2,humidityP);
        // Write to ThingSpeak. There are up to 8 fields in a channel, allowing you to store up to 8
        different
        // pieces of information in a channel. Here, we write to field 1.
        int x = ThingSpeak.writeFields(myChannelNumber, my WriteAPIKey);
        if(x = 200) {
          Serial.println("Channel update successful.");
        }
        else {
          Serial.println("Problem updating channel. HTTP error code " + String(x));
        }
        lastTime = millis();
      }
    }
  }
}

```

- 6a UDP

```

#include <WiFi.h>
#include <WiFiUdp.h>
#include <DHT.h>

```

```

// Replace with your network credentials
const char* ssid = "abcd";
const char* password = "ZXCV@098";
// Replace with your DHT sensor type and pin
#define DHT_PIN 32
#define DHT_TYPE DHT11
// UDP server port
#define UDP_PORT 5000
// Create an instance of the DHT sensor
DHT dht(DHT_PIN, DHT_TYPE);
WiFiUDP udp;
float humidity = 0.0;
void setup() {
  Serial.begin(115200);
  // Connect to Wi-Fi
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Connecting to WiFi...");
  }
  Serial.println("Connected to WiFi");
  // Print the ESP32 IP address
  Serial.print("ESP32 IP address: ");
  Serial.println(WiFi.localIP());
  // Start the UDP server
  udp.begin(UDP_PORT);
  Serial.println("UDP server started");
  // Initialize DHT sensor
  dht.begin();
}
void loop() {
  // Wait for incoming UDP packets
  int packetSize = udp.parsePacket();
  if (packetSize) {
    // Read the incoming packet
    char packetData[packetSize];
    udp.read(packetData, packetSize);
    String request = String(packetData);
    if (request.equals("get_humidity")) {
      // Read humidity from the DHT sensor
      humidity = dht.readHumidity();
      // Send humidity data to the UDP client
      udp.beginPacket(udp.remoteIP(), udp.remotePort());
      udp.printf("Humidity: %.2f%%", humidity);
    }
  }
}

```

```

udp.endPacket();
Serial.printf("Sent humidity data: %.2f%%\n", humidity);
}
}
}

```

- 7a TCP

```

#include <WiFi.h>
#include <WiFiClient.h>
#include <DHT.h>
// Replace with your network credentials
const char* ssid = "abcd";
const char* password = "ZXCV@098";
// Replace with your DHT sensor type and pin number
#define DHT_PIN 32
#define DHT_TYPE DHT11
DHT dht(DHT_PIN, DHT_TYPE);
WiFiServer server(8888); // TCP server port
void setup() {
  Serial.begin(115200);
  delay(10);
  // Connect to Wi-Fi
  Serial.println();
  Serial.print("Connecting to ");
  Serial.println(ssid);
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.print(".");
    Serial.println("");
    Serial.println("WiFi connected");
    Serial.print("IP address: ");
    Serial.println(WiFi.localIP());
    dht.begin();
    server.begin();
  }
  void loop() {
    WiFiClient client = server.available();
    if (client) {
      Serial.println("New client connected");
      // Wait for data from client
      while (client.connected()) {
        if (client.available()) {

```



```
String request = client.readStringUntil('\r');
client.flush();
if (request.indexOf("/humidity") != -1) {
  // Read humidity data from DHT sensor
  float humidity dht.readHumidity();
  if (isnan(humidity)) {
    Serial.println("Failed to read humidity from DHT sensor");
    client.println("Failed to read humidity from DHT sensor");
  } else {
    Serial.print("Humidity: ");
    Serial.print(humidity);
    Serial.println("%");
    client.print("Humidity: ");
    client.print(humidity);
    client.println("%");
  }
}
// Close the connection
client.stop();
Serial.println("Client disconnected");
}
}
}
```